



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

Malaysia-Japan International Institute of Technology (MJIT) · Software Engineering

SCSE1203 – Software Engineering Principles

Dr. Siti Nur Khadijah Aishah Binti Ibrahim

PROJECT CHARTER

Virtual Internship Portal

Group 1 – BOLEH

Group Name	Group 1 – BOLEH
Project Topic	Virtual Internship Portal
Course	SCSE1203 – Software Engineering Principles
Faculty	Malaysia-Japan International Institute of Technology (MJIT), UTM KL
Submission Deadline	[30/04/2026]

SECTION A	Team Details		
No.	Full Name	Matric No.	Scrum Role
1	Nahid Hasan Rayan	A25MJ0186	Product Owner
2	Jobayer Alam	A25MJ0184	Scrum Master
3	Khalid Waleed Issa	A25MJ0185	Development Team
4	Zay Yar Shin	A25MJ0187	Development Team
5	Ghazi Anwar Bin Ahmad Razif	A25MJ5072	Development Team
Role Note: Product Owner — Defines the product vision and maintains the backlog. Scrum Master — Facilitates Agile ceremonies and removes impediments. Development Team — Designs, implements, and tests the product.			

SECTION
B

Project Charter

1. Project Objectives

The Virtual Internship Portal isn't the usual job portals you see cheaply advertised on social media, its an academic service made to fill that gap between what our amazing doctors teach us, and what the professional world wants from us. At this moment, students from everywhere suffer from a very unorganized internship search, from using multiple platforms, apps and websites to do the basics. This is just useless and often ends up in a bad experience for students. Our goal as UTM students is to remove that confusion. We want to replace it with a structured and satisfactory experience.

- For Companies: We want to give the industries a reliable university backed app to find UTM talent. Instead of the messy and inconsistent job posts on every edge of the internet, companies can use workflows to list their requirements, deadlines, and the truth of what they want from the hires.
- For the 3rd and 4th Year students: Our priority is to make the application simple and remove the stress for their future internships. The portal allows them to register, manage their professional profiles and CVS. And track every application in one place rather from going to one website to another.
- For 1st and 2nd Year Students: Team Boleh thinks career preparation should start as early as it can be. Junior students can use and watch over the portal to see the skills needed in the market. For example, which stacks or CGPA are needed to allow the student to prepare for internships before they have a chance to apply.

Primary User Expectation	System Commitment in This Charter
Companies need credible access to qualified student applicants.	Provide verified company profiles and structured internship listing workflows.
Senior students need an efficient application journey.	Our platform focuses on helping seniors track and filter their applications in real-time, so they don't have to jump between different websites.
Junior students need earlier career visibility.	We're including a "Career Radar" view that shows exactly which technical skills and CGPA marks are trending in the current market.
Students need a collaborative knowledge environment.	We'll set up a moderated forum where students/Alumni can swap advice and share their own interview experiences directly.

2. Project Scope

We are defining the scope of our Virtual Internship Portal based on what our team can actually deliver within the SCSE1203 project cycle. Think of this section as our "reality check" it clarifies exactly what we're committed to building and, just as importantly, what is off limits for this semester. By setting these boundaries now, we ensure that our "ambitions" don't outpace the time we actually have in class.

Our primary focus is the core experience for both students and company users. This includes the functional portal, the low-fidelity prototypes, and all the Scrum based reporting we need for our final evaluation. We are intentionally avoiding features that would add massive technical complexity or require us to provide long term support after the course ends. If a new request pops up, we'll add it to the backlog for later rather than trying to squeeze it into this charter.

Boundary Area	Included in Scope	Explicitly Out of Scope
Student-facing functions	Registration, profile management, internship search, filtering, application submission, and application tracking.	Automated AI matching, interview scheduling, or full career-services automation.
Company-facing functions	Company profile creation, listing publication, vacancy updates, and visibility of applicants through a structured workflow.	Enterprise recruitment integrations or advanced screening automation.
Community features	Discussion forum for questions, advice, and peer knowledge exchange.	Real time private messaging or live social collaboration tools.
Market-awareness features	Display of frequently requested skills, CGPA expectations, and trends derived from internship listings.	Predictive analytics or production-grade labour-market intelligence services.
Platform targets	Web-based portal and low-fidelity prototype covering major flows.	Native mobile applications or multi-platform commercial release.
Operational responsibility	Course-period development, testing, and documentation.	Long-term hosting, production deployment, or post-course maintenance obligations.

We might propose changes during our backlog refinement or sprint reviews, but we'll only commit to them if they fit our team's capacity and meet Dr. Aishah's expectations. This approach protects us from "scope creep" while still letting us adapt as we learn more through our Scrum sprints.

3. Analysis Methods

We're starting to take a small approach to gather requirements, using validation from students that fits our workflow. Instead of making a set of rules to follow from the start. The charter will let us gather information on the go, and hopefully we can turn that info into user stories and check back through review cycles to make sure the portal does what we wanted it to as we evolve.

At first, our data collection is going to be based on user surveys, a skill we picked up from our probability and statistics class. Just specific questions and direct talks with other students or company representatives if we get the chance. This helps us see past the “cool features” and find out the basics that need to be built and gaps in the information we have. We aren't looking for a feature list, just to understand how things like trust and academic prep change throughout internship plans.

Validation is a big process for us, we'll use low prototypes, and easy walkthroughs and our daily standups to keep testing our ideas. This will make sure we stay flexible while the surveys we do tell us what's needed. Prototypes let us test our assumptions, and sprint reviews give us direct feedback from lecturers and our friends. By the end of the project, every decision we make for the portal will be backed.

Method	Purpose Within the Charter	Expected Output
User surveys	Capture needs, frustrations, and preferred features from student and stakeholder perspectives.	Structured feedback that informs initial requirements and priorities.
Stakeholder discussion	Clarify expectations from lecturer, team members, and potential company side users.	Shared understanding of project goals, boundaries, and decision criteria.
User story mapping	Translate user needs into backlog ready functional statements.	Prioritised user stories linked to end-user value.
Low-fidelity prototyping	Test navigation and feature assumptions before heavy implementation effort.	Prototype feedback and revised interface decisions.
Sprint reviews	Evaluate increments and capture improvement actions at the end of each sprint.	Lecturer and peer comments that influence backlog refinement.
Daily standups	Surface blockers, progress updates, and coordination needs during development.	Operational visibility and timely issue escalation.

4. Key Participants & Time Commitment

Our projects success is dependant on three groups, the students, Dr Aishah as our evaluator, and a small group of representative users who will give us feedback during our analysis and review phases. We've deliverately separated the roles of whos working making the portal from whos checking our academic work or providing real world insight. This is crucial because we need people working within a university setting at MJIT rather than meeting the commercial contract. So we end up delivering learning and the needs.

When it comes to time, think of the numbers as our best estimates for the remaster rather than a legal schedule as were just students. Our team consisting of Nahid, Khalid, Shin, Ghazi, and Jobayer will do all the big lifting throughout the project. Meanwhile Dr Aishah's time is focused on teaching us and giving us the learning we need. As for the users, they aren't involved daily although they'll help with surveys and quick walkthroughs when we need their opinion.

Participant	Role	Time Commitment / Involvement
Nahid Hasan Rayan	Product Owner	Approximately 4-6 hours per week for product vision alignment, backlog prioritisation, and stakeholder coordination.
Jobayer Alam	Scrum Master	Approximately 3-5 hours per week to facilitate standups, planning, reviews, and impediment management.
Ghazi Anwar Bin Ahmad Razif	Development Team	Approximately 5-8 hours per week for analysis, design, implementation, testing, and supporting documentation.
Khalid Waleed Issa	Development Team	Approximately 5-8 hours per week for analysis, design, implementation, testing, and supporting documentation.
Zay Yar Shin	Development Team	Approximately 5-8 hours per week for analysis, design, implementation, testing, and supporting documentation.
Dr. Siti Nur Khadijah Aishah Binti Ibrahim	Lecturer / Evaluator	Participates at class checkpoints, sprint reviews, and formal assessment moments across the course.
Representative Senior Students	Primary End User Participants	Contribute survey responses and prototype feedback focused on application and tracking workflows.
Representative Junior Students	Secondary End User Participants	Contribute feedback on awareness features, skills visibility, and perceived usefulness of planning support.
Company Representatives	Consulted Stakeholders	Provide limited perspective on company profile content and internship listing expectations when accessible.

In our team setup, Nahid takes charge of the priority list while Jobayer focus on keeping our workflow steady and removing blockers. Ghazi, Khalid, and Shin handle the heavy lifting of development and testing. Dr. Aishah is there to grade our progress and academic depth, while our student users help us figure out if the portal is actually usable. We chose this structure to keep our decision making clear and simple throughout the semester

5. Project Deliverables

We're delivering more than just a portal, we want to show the paper trail of how we got here, since this is a project at MJIIT, we have to prove that we followed all the real world engineering habits. By planning, sprint management and reflect.

Every item on our list helps show that we managed this project responsibly and made smart engineering calls. Our documentation keeps us accountable for what we need and the prototypes and the final portal will prove that our idea is something that students can benefit from.

Finishing this semester isn't just about handing over screenshots or files. We want to make sure our project tells a story, that we've stayed to our scope and managed our backlog properly. We built a portal that meets the expectations of our users. The table below outlines the main objectives we want to give during this course.

Category	Deliverable	Purpose Within the Project
Governance	Project Charter	Defines authority, scope, responsibilities, and overall project direction.
Requirements	User Stories and Product Backlog	Translate user needs into prioritised, reviewable backlog items.
Scrum execution	Sprint Backlogs	Record sprint-level commitments and planned implementation focus.
Scrum reporting	Daily Standup Records	Provide evidence of team coordination, issue tracking, and work visibility.
Iteration review	Sprint Review and Retrospective Records	Show progress demonstration, feedback capture, and process improvement actions.
Progress tracking	Burndown or Burn up Charts	Reflect work rate, remaining effort, and scope movement across sprints.
Design artefact	Low Fidelity Prototype	Validate user flow, layout logic, and navigation assumptions before or alongside implementation.
Software product	Virtual Internship Portal	Provide the functional web-based outcome defined by this charter.
User support	Onboarding guidance and concise help content	Support first-time use and reduce training overhead for target users.

6. Evaluation

We are evaluating the virtual internship portal with the reviews of our lecturer, feedback during our sprints and just basic tests from the users. If we look at it from an academic perspective, our success isn't just about whether the code works or not. It's about following the proper engineering process, staying true about our progress and that we can justify every design choice or management choice we made.

Dr. Aishah, she's going to be our lead evaluator as mentioned previously checking our work against and seeing if it matches the course requirement. We're going to use our sprint reviews as the primary way to get good feedback and make good improvements before our final submission. We also want to look for inputs from our peers' opinions, walkthroughs and introductions of our prototypes and our own team's perspective to make sure the portal can serve everyone.

For this project, a good evaluation would mean that everything is lined up, our original goals, the scope we set. Our iterative work and our final results. That's why we want the portal and all the supporting documents to be judged together.

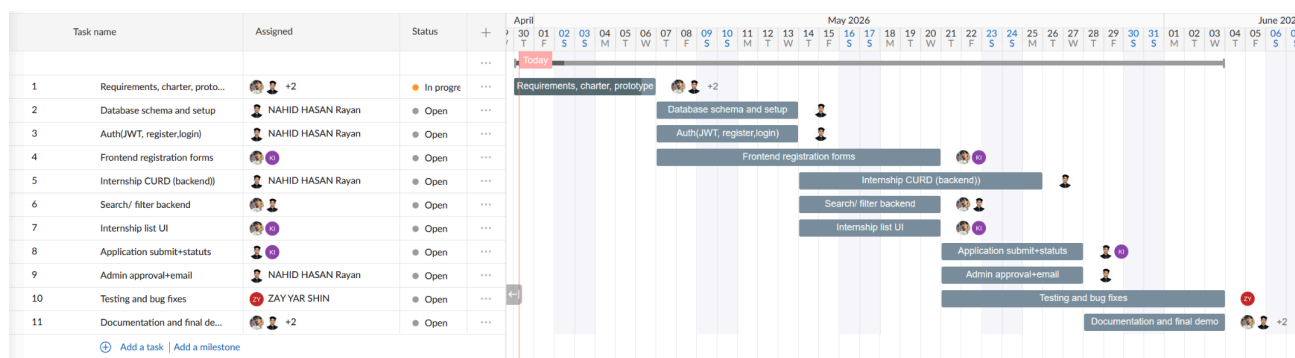
Evaluator or Source	Method of Evaluation	Evidence Expected
Lecturer	Review of charter, sprint outputs, prototype, and final project artefacts.	Assessment feedback, grading outcomes, and direction for refinement.
Sprint reviews	Demonstration of completed increments at the end of each sprint.	Observed progress, feedback notes, and backlog adjustment actions.
Peer reviewers	Informal comments on usability, clarity, and perceived usefulness.	Practical observations that may highlight interface or workflow issues.
Representative users	Prototype walkthroughs and feature oriented feedback sessions.	Qualitative insight into whether the portal supports real user expectations.
Team retrospective	Internal reflection on process effectiveness and blockers.	Action points for improving collaboration, planning, and delivery quality.

7. Timeline & Reporting Milestones

We're running this project on a five week agile scrum rhythm during this 2026 semester. Each week has a clear focus from getting the foundations right in week one, through two sprint cycles, to a final consolidation week for testing and the demo. We're using project weeks rather than specific dates in this charter in case we need to stay flexible for last minute class schedule shifts.

Instead of just a final hand in, we want to treat the timeline as a constant reporting loop. We'll do daily standups for quick internal check ins, and formal sprint reviews to show Dr. Aishah our progress at key milestones. After each sprint, we'll also run retrospective sessions to look at what we learned and fix any issues before the next week. This way, the schedule wouldn't just be a deadline it'll be a framework that keeps us all on the same page.

Phase	Period	Milestone / Deliverable
Project Week 1	Initiation and chartering	Finalise requirements, project charter, and prototype baseline. All team members aligned on scope and responsibilities.
Project Week 2	Foundation and setup	Database schema and setup, authentication (JWT, register, login), and frontend registration forms begin.
Project Week 3	Sprint 1 execution	Internship CRUD (backend), search and filter backend, and internship list UI delivered. Frontend registration forms completed.
Project Week 4	Sprint 2 execution	Application submission and status tracking implemented. Admin approval and email notifications completed. Testing and bug fixes begin.
Project Week 5	Consolidation and submission readiness	Complete testing and bug fixes, finalise documentation, and deliver the final demo.



Pictured: Gantt Chart Made with GanttPro. Link to chart will be given in appendix of project proposal

We might adjust task assignments within a week if the team agrees through our scrum process, but we don't plan on relaxing our reporting habits. For every week in this timeline, we want to make sure there is a clear deliverable the team can point to not just work in progress, but something reviewable. The five week structure is tight, so staying on top of it from week one is non negotiable.

8. Training

We aren't planning on running any training sessions for this portal. Instead we want to build the training right into the design itself. Our main goal is for the interface to be intuitive enough that anyone can figure out the basics like setting up a profile, browsing internships, and posting on the forms.

If someone does get stuck, we have built in support tools ready. Quick cues, short tooltips and clear help text. This works best for us because it cuts the need for face to face training while proving we've went over want needs to make the site easy to use for our peers.

Khalid, Shin, and Ghazi from our development team are responsible for creating and keeping the support and help material updated. Nahid will check in as a product owner to make sure that any guidance we offer lines up with what users usually ned. In the charter training, it just means making sure our users are given all the resources they need.

Training Support Component	Description	Responsible Party
Intuitive interface design	Use familiar navigation patterns and clear task flows to reduce onboarding effort.	Development Team
Onboarding prompts	Provide first-use guidance for key actions such as registration, profile completion, and internship browsing.	Development Team
Contextual tooltips	Clarify fields, labels, and features that may not be immediately obvious to first-time users.	Development Team
Concise help content	Offer short written guidance that explains the primary user journeys in the portal.	Development Team with Product Owner review

9. Maintenance

Ghazi, Khalid, and Shun from our dev team are handling maintenance while we stay inside our timeline and backlog. To us, maintenance is about squashing bugs, making the site a bit easier to use, and updating our docs whenever Dr. Aishah or our own reviews highlight something that needs fixing.

We want to stick to a sprint based maintenance model, rather than something that doesn't end. Whenever we find an issue during testing or get feedback from a peer. We'll log it in the backlogs so we can prioritize it alongside our other work. This makes sure that everything is organized and prevents the project from spiraling. It'll also let us improve the portal using iteration by iteration. We also want to use version control so we can see who exactly changed what and so it can provide us with backups incase anything bad happens.

It is just as important to be clear about where this ends our charter does not cover indefinite support once the course is finished. If we were to keep the portal running after our final submission, we would need a totally different set of resources and a new agreement. For now, our maintenance responsibilities will end exactly when the project lifecycle for the semester does

Maintenance Area	Owner	Approach	Boundary
Defect correction	Development Team	Record issues during testing or reviews and resolve them in the next feasible sprint item.	Within approved course timeline only.
Usability refinement	Development Team with Product Owner input	Translate recurring usability concerns into prioritised backlog improvements.	Limited to in scope changes.
Documentation upkeep	Scrum team	Update charter aligned artefacts when decisions or scope interpretations change.	Must remain consistent with lecturer expectations.
Version control and traceability	Scrum team	Maintain shared repository history for collaborative work, rollback support, and accountability.	Applies during project development lifecycle.

**SECTION
C****Key Assumptions and Constraints**

Here, we've put together these assumptions and constraints to help explain how we're using this charter. They highlight the needs and conditions we need to deliver the project and practical limits that we'll use to keep our scope and speed up. Listing these out helps show why we're managing this as an academic portal for our course instead of making it a full commercial platform.

Type	Item	Charter Position
Assumption	Representative students will be available for light feedback activities.	The team assumes at least limited access to student opinions through surveys, walkthroughs, or informal review.
Assumption	The lecturer will provide milestone based academic guidance.	Sprint reviews and assessment checkpoints are assumed to remain the main source of formal project evaluation.
Assumption	The team can collaborate through shared tools and version control.	Project artefacts and source materials are assumed to be maintainable in a common repository or shared workspace.
Assumption	A web based implementation is sufficient for the course objective.	The charter assumes that a browser-accessible portal adequately demonstrates the intended solution concept.
Constraint	Time is limited by the semester schedule.	Feature selection and scope depth must remain realistic within the academic delivery window.
Constraint	The project is assessed in an educational setting.	Documentation, process discipline, and demonstrable learning outcomes are mandatory alongside software output.
Constraint	Access to companies may be limited.	Company side requirements may rely partly on inferred needs and secondary validation rather than extensive stakeholder involvement.
Constraint	Long term deployment is outside the project mandate.	The team is not obligated to provide production hosting, post course support, or operational continuity.

**SECTION
D****Communication and Reporting Plan**

Since we're running this project with Scrum, we're making communication a regular part of how we manage things instead of just an afterthought. The plan we've laid out below shows how we'll share information between our team, Dr. Aishah, and our users. This is how we'll make sure our decisions, progress, and any issues stay clear and visible for the whole semester.

Audience	Channel or Format	Frequency	Purpose
Scrum team	Daily standup or short coordination update	Frequent / working days	Share progress, surface blockers, and coordinate immediate tasks.
Scrum team	Sprint planning meeting	At the start of each sprint	Confirm sprint goals, responsibilities, and backlog focus.
Lecturer / evaluator	Sprint review demonstration	At the end of each sprint	Show working progress and receive formal formative feedback.
Scrum team	Sprint retrospective	After each sprint review	Reflect on process effectiveness and define improvement actions.
Representative users	Survey or prototype walkthrough	When needed during analysis and validation	Collect insight on usability, expectations, and clarity of features.

**SECTION
E****Project Risks and Mitigation**

We added this risk section to show the though we put into things that can threaten our quality and project control. We don't need a full on corporate level risk register but we need to call out the main problems that hit academic projects at this size. We also included a plan for how to handle these issues if they happened.

Risk	Potential Impact	Planned Mitigation	Owner
Scope creep	The team may commit to more features than the semester allows.	Use backlog prioritisation and lecturer aligned scope control before accepting additions.	Product Owner and Scrum Master
Uneven team workload	Delivery quality may suffer if tasks are not distributed or tracked well.	Use sprint planning, standups, and visible task ownership to rebalance work early.	Scrum Master and Development Team
Limited user feedback	Features may reflect assumptions more than validated need.	Use at least lightweight surveys and walkthroughs, and record uncertainty in design decisions.	Product Owner
Time pressure near submission	Documentation and testing quality may decline late in the semester.	Stage deliverables across sprints and maintain artefacts continuously instead of at the end only.	Entire Scrum Team

Risk	Potential Impact	Planned Mitigation	Owner
Technical defects or rework	Unexpected issues may consume planned sprint effort.	Keep increments small, test progressively, and log corrective tasks through the backlog.	Development Team
Misalignment with evaluation expectations	The final output may be penalised despite technical effort.	Use sprint reviews and periodic lecturer feedback to confirm that process and artefacts remain aligned.	Product Owner and lecturer checkpoints

SECTION F

Approval & Acknowledgement

By proceeding with this project, all team members acknowledge their roles, responsibilities, and commitment to the deliverables outlined in this charter.

Nahid Hasan Rayan
Product Owner

Jobayer Alam
Scrum Master

Dr. Siti Nur Khadijah Aishah Binti Ibrahim
Lecturer / Evaluator

Ghazi Anwar Bin Ahmad Razif
Development Team

Khalid Waleed Issa
Development Team

Zay Yar Shin
Development Team